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**SUMMER TRAINING / INTERNSHIP REPORT ON
“AUTOMATED WATER PURIFICATION WITH PLC
AUTOMATION”**

(9th June 2025 to 31th July 2025)

Submitted in partial fulfilment of the requirement for the award of the degree of
BACHELOR OF TECHNOLOGY

SCHOOL OF ELECTRONICS AND ELECTRICAL ENGINEERING

Submitted By:

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Registration No.: 12507019

Under the Guidance of:

DR. HARPREET SINGH BEDI



**LOVELY PROFESSIONAL UNIVERSITY
PHAGWARA, PUNJAB
2025 – 2026**

DECLARATION

I, Ette Pruthvi Teja Reddy, Registration No. 12507019, student of Bachelor of Technology at Lovely Professional University, Phagwara, hereby declare that the Summer Training/Internship Report entitled “Automated Water Purification With PLC Automation”, submitted in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology, is an authentic record of my own work carried out during the period from 9th June 2025 to 31th July 2025 under the guidance of Dr. Harpreet Singh Bedi.

The matter presented in this report has not been submitted by me, in part or in full, for the award of any other degree or diploma of this or any other University or Institute. Wherever contributions of others are involved, every effort has been made to indicate this clearly, with due reference to the literature and acknowledgement of collaborative research and discussions.

Date:

Place: Phagwara

(Ette Pruthvi Teja Reddy)

Registration No.: 12507019

CERTIFICATE

This is to certify that the Summer Training/Internship Report entitled “Automated Water Purification With PLC Automation” submitted by ETTE PRUTHVI TEJA REDDY(Registration No. 12507019) in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology to Lovely Professional University, Phagwara, is a bona fide record of the training/internship carried out by him during the period from 9th June 2025 to 31th July 2025 under my guidance and supervision.

To the best of my knowledge, the matter embodied in this report has not been submitted elsewhere for the award of any other degree, and the training activities and project work described herein have been carried out satisfactorily.

Date:

(Dr. Harpreet Singh Bedi)
Faculty Guide Lovely Professional
University, Phagwara

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Lovely Professional University for providing me the opportunity to undertake this summer training/internship on “Intelligent Industrial Automation: AI, Sensors, Actuators & PLC Integration”. I am deeply thankful to my faculty guide, Dr. Harpreet Singh Bedi, for his valuable guidance, constant encouragement, and constructive feedback throughout the duration of this training, which greatly contributed to the successful completion of this report and the associated project work.

I extend my sincere thanks to the Division of Research & Development and the faculty and staff of the Industrial Automation Laboratory for providing access to the PLC trainer kits, Logix Pro, and Twindosvite software used during this training, and for their support in clarifying practical concepts related to industrial process automation.

I am also grateful to my family and friends for their continuous support, patience, and motivation throughout this training period. Finally, I would like to thank all those who, directly or indirectly, helped me during the course of this training and in the preparation of this report.

(Ette Pruthvi Teja Reddy)

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CHAPTER 1: INTRODUCTION OF ORGANIZATION

1.1 ABOUT LOVELY PROFESSIONAL UNIVERSITY

Lovely Professional University (LPU) is one of India's largest private multidisciplinary universities, established in 2005 and located at Phagwara in the Kapurthala district of Punjab. The University offers undergraduate, postgraduate, and doctoral programmes across streams such as engineering and technology, management, computer applications, agriculture, pharmacy, law, and the sciences. LPU is accredited by NAAC with an A++ grade and is recognised for its emphasis on industry-integrated learning, practical exposure, and skill development alongside conventional classroom instruction.

As part of its academic model, LPU requires undergraduate students to undertake a Summer Training/Internship between academic years so that classroom learning in areas such as programming, electronics, instrumentation, and automation can be reinforced through hands-on exposure to real or simulated industrial systems. This report is prepared in fulfilment of that requirement for the Bachelor of Technology programme, Registration No. 12507019, for the academic session 2025–2026.

1.2 TRAINING FACILITY – INDUSTRIAL AUTOMATION LABORATORY

The present summer training was undertaken at the Industrial Automation and PLC Training Laboratory functioning under the Division of Research & Development at Lovely Professional University. This laboratory is equipped with Programmable Logic Controller (PLC) trainer kits (Siemens S7-1200 series), Human Machine Interface (HMI) panels, Logix Pro workstations, and a range of sensor and actuator modules such as thermocouples, proximity sensors, load cells, solenoid valves, and variable frequency drives.

The laboratory is designed to let students design, simulate, and test control logic for processes that closely resemble those found in real process industries – including batching, thermal regulation, material handling, and packaging – using industry standard programming environments such as Siemens TIA Portal, WinCC SCADA, and the Factory IO 3D simulation platform. For this training, the glass product manufacturing process was adopted as the reference case study, since it involves a representative sequence of automation-intensive stages (batching, melting/furnace

control, forming, annealing, and inspection) that are well suited to demonstrating PLC based control concepts.

1.3 OBJECTIVE OF THE TRAINING CELL

The broad objective of the Division of Research & Development in offering this training is to bridge the gap between theoretical curriculum and industrial practice by exposing students to the type of control hardware, programming languages, and monitoring systems that are actually deployed on modern manufacturing shop floors. Students are guided to understand a manufacturing process end-to-end, identify the subprocesses that benefit most from automation, and then design, program, and validate a working control solution for a chosen sub-process, which subsequently forms the basis of the individual internship project.

CHAPTER 2: SUMMER TRAINING COURSE / INTERNSHIP CONTENT DETAIL

The summer training was carried out over a period of eight weeks, from 9th June 2025 to 31st July 2025. The training combined theoretical sessions on the PLC fundamentals with hands-on laboratory exercises using PLC trainer kits, HMI panels, and simulation software. The week-wise breakup of the training content is summarised in Table 1.

Table 1: Week-wise Summer Training Schedule

Week	Topic Covered
Week 1	Introduction to various measurement systems; PLC operational faults; Introduction to sensors and actuators in automated systems; Basic fault-finding methodologies.
Week 2	The central processing unit; Process scanning considerations; Memory organization in industrial controllers; I/O module selection and wiring criteria.
Week 3	Flowcharting as a programming method; PLC counters, indicators, and others; Basics of HMI (Human-Machine Interface) screen design; State machine logic design principles.
Week 4	Contact and coil I/O programming examples; A look at fail-safe circuits; Industrial safety standards and emergency stop configurations; Integration of automated relay logic.
Week 5	Large process ladder diagram construction; Examples of counter function industrial applications; Modular programming structures; Coordinating robotic arm movements and interlocking using ladder logic.
Week 6	Examples of timer function industrial applications; Hierarchy of Artificial Intelligence, Machine Learning, and Deep Learning; Industrial data collection strategies using Python scripting; Basics of SCADA architectures
Week 7	Supervised, unsupervised, and reinforcement learning paradigms; Real industry use-cases in predictive maintenance, vision inspection, and process optimization; Deploying machine learning models for industrial anomaly detection.
Week 8	Advanced programming languages: IEC 1131, other programming languages, applications of PLC; Integrating open-source software frameworks for Industrial IoT (IIoT); Final project testing and documentation.

2.1 PLC PROGRAMMING FUNDAMENTALS COVERED

During the laboratory sessions, emphasis was placed on developing a working understanding of Ladder Diagram (LD) programming, the most widely used PLC programming language on the shop floor. Topics covered included digital and analog input/output addressing, timers (TON/TOF) and counters (CTU/CTD), latching and interlocking logic, analog signal scaling for sensors such as RTDs and load cells, and an introduction to Function Block-based PID control for closed-loop temperature regulation.

2.2 SOFTWARE AND TOOLS USED DURING TRAINING

- **LogixPro** – Utilized as the primary integrated environment for developing, simulating, testing, and troubleshooting PLC ladder logic. The software provided interactive, animated simulations (such as batch mixing, conveyor systems, and traffic control) that closely mirrored real-world industrial environments. This enabled comprehensive hands-on practice with relay logic, timers, counters, and data handling instructions, allowing for immediate feedback on program execution without the need for physical hardware.
- **Twindosvite Software** – Deployed for advanced process visualization, operational monitoring, and system diagnostics. It served as a dynamic platform to test the developed control logic against complex, simulated industrial processes. This software facilitated real-time observation of system responses, graphical interface interaction, and the rigorous validation of safety interlocks and control strategies before any theoretical field deployment.

2.3 SKILLS GAINED

Through the intensive eight-week programme, a comprehensive skill set in industrial automation programming and system design was developed. Key proficiencies include:

- **System Analysis & Logic Design:** The ability to accurately read and interpret complex process and instrumentation flow diagrams (P&ID) and translate those requirements into efficient, well-structured ladder logic routines using software simulators.
- **Virtual Configuration & Addressing:** Practical experience in systematically configuring virtual input/output modules, assigning memory addresses, and managing internal data structures to ensure seamless communication between the simulated PLC and the process model.
- **Troubleshooting & Optimization:** Enhanced logical problem-solving abilities, specifically in monitoring live program execution, isolating logic faults, and optimizing automated process flows under simulated stress conditions.

CHAPTER 3: SUMMER TRAINING / INTERNSHIP PROJECT DETAIL

3.1 INTRODUCTION

Water purification is a critical continuous process where precise control of pumps, valves, and sterilization equipment is essential to ensure safe drinking water and prevent hardware failure. In traditional, manually operated water treatment plants, the reliance on human intervention for operating valves and switches often leads to inconsistent processing, potential overflow hazards, and equipment damage. The project undertaken during this training addressed the need to design and simulate a Programmable Logic Controller (PLC) based automated water purification system. The objective was to replace unreliable manual operations with a fully automated, closed-loop ladder logic workflow that governs feed pumps, UV sterilization timing, and safety interlocks, ensuring efficient, error-free water treatment with zero human intervention during standard operation.

3.2 PROBLEM STATEMENT

Traditional water purification systems suffer from several operational inefficiencies and safety risks due to their reliance on manual control. Manual operation of valves and switches frequently leads to human error, resulting in inconsistent water processing and flow rates. A major critical issue is hardware damage; for instance, feed pumps running dry when source tanks are empty cause frequent motor burnouts. Furthermore, contamination risks arise if the Ultraviolet (UV) sterilization lamps are not properly timed, allowing unsterilized water to contaminate the clean holding tank. Additionally, the lack of automated float sensor interlocks leads to wasted pure water and facility flooding due to overflow hazards. This project aimed to solve these issues by implementing a robust PLC ladder logic system to automate safety checks, pump activation, and sterilization delays..

3.3 WORKING

The automated water purification system developed during this project operates as a continuous, PLC-supervised sequence utilizing detailed ladder logic. The step-by-step working of the system is summarized below:

- The operator initiates the purification cycle by pressing the START push-button (I:1/0), which energizes the master System Run bit (B3:0/0). This bit is placed in parallel to create a "seal-in" latch circuit.

- The system constantly monitors safety interlocks. The Stop, Emergency Stop, and Pure Tank Full sensors are wired in series as normally closed (NC) contacts. If any safety condition is met, the circuit breaks, safely halting the system.
- To ensure dry-run protection, the system checks tank levels before pump activation. The Feed Water Pump (O:2/0) will only turn on if the Raw Tank Low Level sensor indicates water is available and the Pure Tank Full sensor indicates there is receiving capacity.
- The activation of the Feed Water Pump directly opens the Solenoid Valve (O:2/1), allowing raw water to flow into the piping system. Simultaneously, a Timer On Delay (TON) triggers a 5-second buffer delay before engaging the UV sterilization lamp.
- Once the 5-second delay timer finishes, its Done Bit goes high, automatically turning on the UV Lamp (O:2/2) to begin sterilization, which immediately starts the main Purification Time tracker.
- During operation, if the Raw Tank Low Level sensor ever goes high (tank empty), an Alarm Buzzer (O:2/3) sounds to alert the operator.
- When the main purification timer reaches its preset value, the cycle completes, and a Purification Completion Indicator (O:2/4) illuminates on the control panel.

3.4 PROJECT OUTCOMES

The following outcomes were achieved as part of the mini-project developed during the training:

- An automatic purification sequence was successfully developed in ladder logic, integrating essential safety interlocks, pump control, and sequenced timer functions for UV sterilization.
- A robust fail-safe mechanism was implemented using float sensor inputs to prevent dry-running of feed pumps and to eliminate tank overflow hazards.
- Precise timing controls were established using TON (Timer On Delay) instructions to ensure a strict 5-second buffer before UV lamp activation, guaranteeing consistent sterilization.
- The complete ladder logic was simulated, tested, and debugged using the LogixPro simulation environment. In testing, the system accurately halted during simulated emergency stops and correctly triggered alarms for empty raw tank conditions.

- Comprehensive I/O allocation mapping and a functional process flow chart were prepared as project documentation deliverables.

3.5 ADVANTAGES

The PLC-based automation approach adopted in this project offers several advantages over the manual Operations:

- **High Reliability:** The PLC-driven logic guarantees consistent operation, offering years of continuous processing without the failures associated with manual switching.
- **Fail-Safe Operations:** Integrated hardware and software interlocks prevent catastrophic failures, completely eliminating tank overflows and dry pump running.
- **Consistent Sterilization:** Logic-enforced timers ensure that water processing and UV exposure parameters are identical in every single cycle, maximizing water purity.
- **Scalability:** The ladder logic structure can be easily expanded in the future to include additional features like Reverse Osmosis (RO) membrane flushes or multiple holding tanks without extensive rewiring.
- **Improved Safety and Efficiency:** By removing the need for constant human oversight, the system optimizes water flow, reduces mechanical wear on pumps, and ensures operator safety.

3.6 DISADVANTAGES

At the same time, the automation approach studied in this project has certain limitations:

- **High Initial Capital Cost:** Implementing industrial PLCs, automated solenoid valves, float sensors, and heavy-duty contactors requires a significantly higher initial investment than simple manual valves.
- **Skilled Personnel Required:** Modifying the system parameters, expanding the logic, or troubleshooting software faults requires technicians trained specifically in ladder logic and automation.
- **Hardware Dependency:** The software logic is strictly dependent on the physical sensors. If a mechanical float switch gets stuck or fails, the PLC will act on false data, potentially causing system halts or overrides.
- **As implemented here,** the system was validated primarily in a simulated environment (LogixPro/Twindosvite); real-world fluid dynamics, pipe

pressure variations, and sensor degradation over time were not fully captured and would require field calibration before actual deployment.

3.7 CHALLENGES FACED

A few practical challenges were encountered while developing and testing the control logic during the training:

- Designing the "seal-in" latch circuits to correctly reset the entire process sequence when an emergency stop or tank-full condition was triggered, ensuring no pumps remained active.
- Accurately synchronizing the Timer On Delay (TON) instructions to ensure the solenoid valve and UV lamp operated in the exact required sequence without race conditions.
- Creating a realistic simulation of water flow and tank filling/draining speeds within the Twindosvite and LogixPro software to properly test the float sensor logic.
- Structuring the ladder logic to clearly separate alarm triggers (like the empty tank buzzer) from standard operational stops (like the pure tank full indicator)

3.8 TECHNOLOGIES USED

The project made use of the following hardware, software, and communication technologies:

- Simulation & Programming Software: LogixPro was used as the primary integrated environment for writing, simulating, and debugging the PLC ladder logic.
- Process Visualization: Twindosvite software was utilized for monitoring system responses and visually testing the control logic against simulated industrial water processes.
- Simulated Hardware & Actuators: PLC CPU logic, internal plumbing and filter routing, heavy-duty contactors and relays for feed water pumps, and solenoid valves for flow control.
- Simulated Field Sensors: Normally Open (NO) and Normally Closed (NC) pushbuttons for Start/Stop/E-Stop commands, and float sensors (limit switches) for monitoring Raw Tank Low and Pure Tank Full levels.

CHAPTER 4: SOFTWARE STIMULATION / OUTPUT

This chapter presents the comprehensive system-level architecture, the detailed input/output (I/O) allocation list, and a representative extract of the ladder logic developed for the automated water purification project described in Chapter 3.

To provide a complete technical overview of the implemented control strategy, the system documentation is divided into the following key areas:

4.1 System Architecture and Process Flow

This section illustrates the physical and logical flow of the water purification sequence. It visually maps the journey of the water from the raw intake tank, through the PLC-controlled feed pumps and solenoid valves, into the timed UV sterilization chamber, and finally to the pure water holding tank. The architecture diagrams highlight the critical touchpoints where the PLC interacts with simulated field devices to maintain closed-loop control.

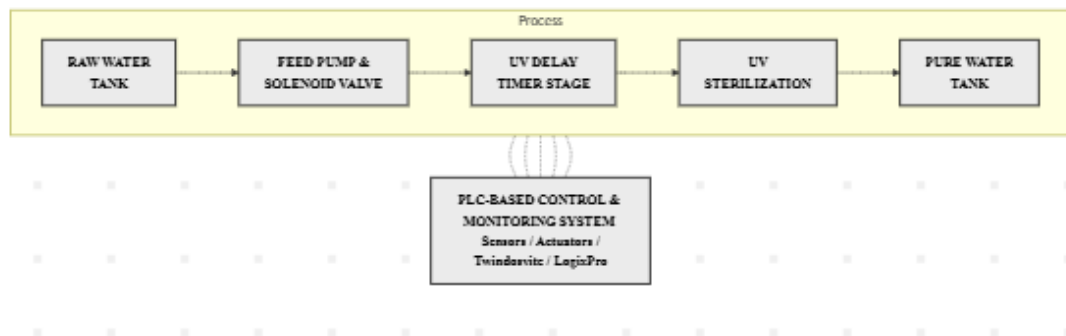


Figure 1: Water Purification with PLC-based Automation Touchpoints

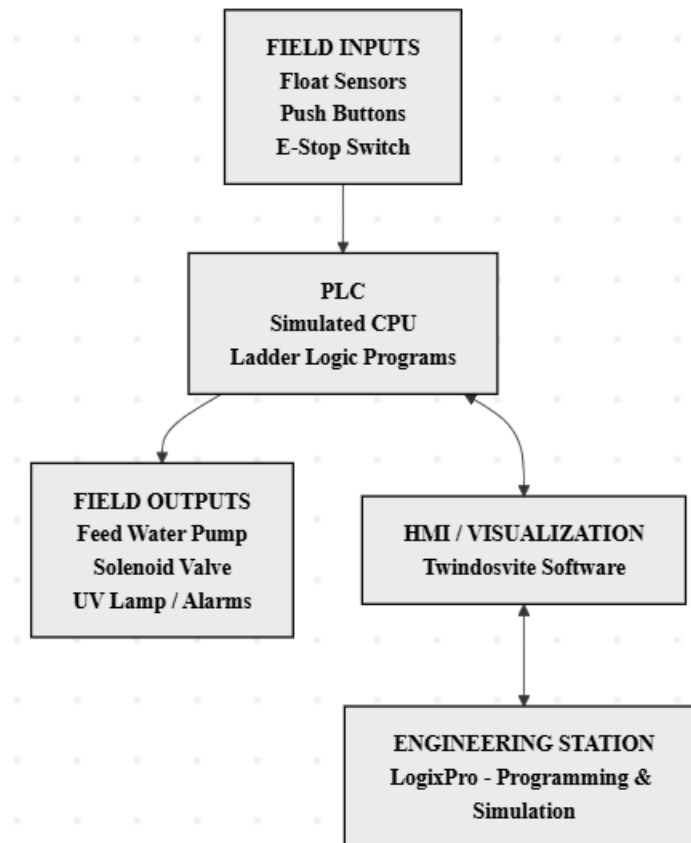


Figure 2: PLC Control System Architecture

4.2 Input / Output (I/O) Allocation

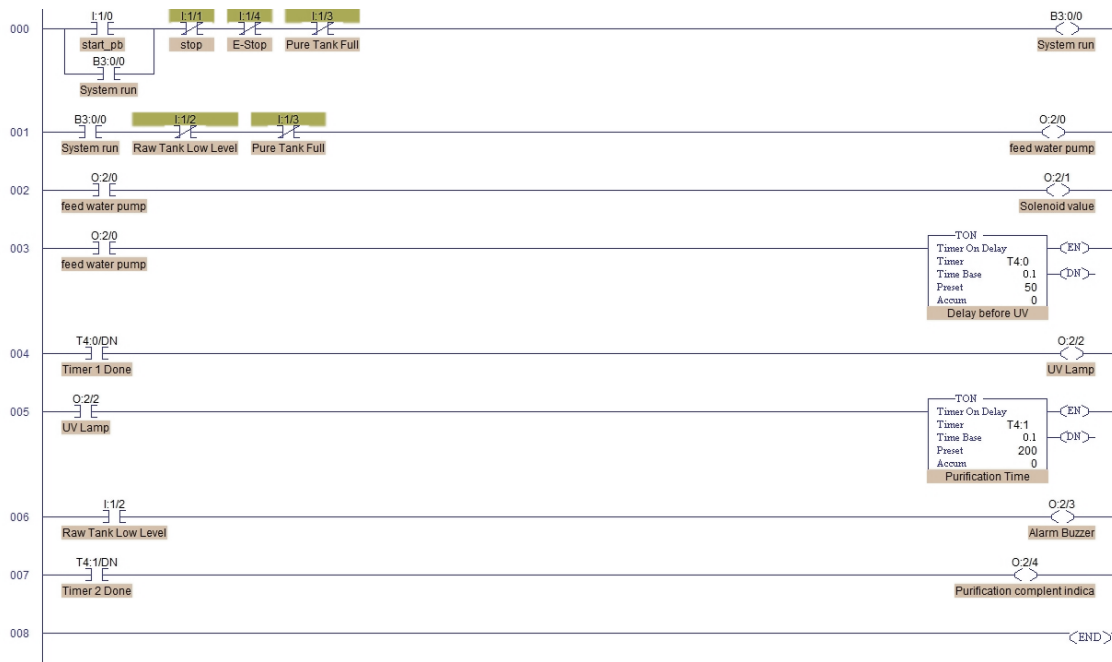
A foundational step in PLC programming is the systematic assignment of physical or simulated field devices to exact controller memory addresses. The I/O list provided in this chapter details the complete mapping strategy used in the LogixPro environment. It clearly defines the digital inputs (such as start/stop pushbuttons and tank level float sensors) and the digital outputs (such as pump motor contactors, solenoid valves, and UV lamp relays) required to execute the purification cycle.

Table 2: Input / Output (I/O) List for the Batching and Furnace Control Logic

Tag Name	Address	Description	Type
start_pb	I:1/0	Operator start push-button	Digital In
stop	I:1/1	Operator stop push-button	Digital In
Raw Tank Low Level	I:1/2	Raw water tank low level float sensor	Digital In
Pure Tank Full	I:1/3	Pure water tank full level float sensor	Digital In
E-Stop	I:1/4	Emergency stop push-button	Digital In
feed water pump	O:2/0	Feed water pump motor control	Digital Out
Solenoid valve	O:2/1	Solenoid valve for raw water flow	Digital Out
UV Lamp	O:2/2	UV sterilization lamp power relay	Digital Out
Alarm Buzzer	O:2/3	Low raw water alarm buzzer	Digital Out
Purification complent indica	O:2/4	Purification completion indicator light	Digital Out

4.3 Ladder Logic Control

This section showcases the core decision-making routines of the automation software. The selected logic extracts highlight the most critical safety and operational rungs of the program, including:



4.4 Boolean Logic Representation

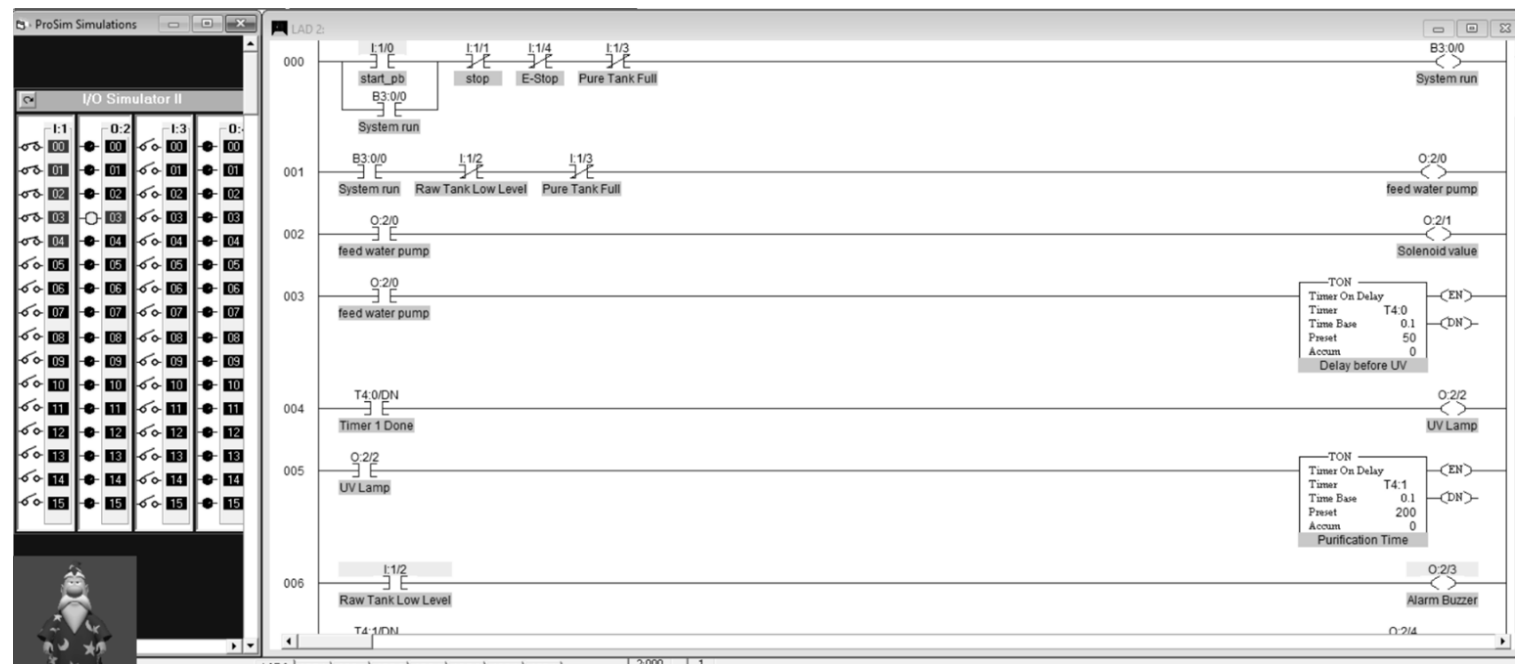
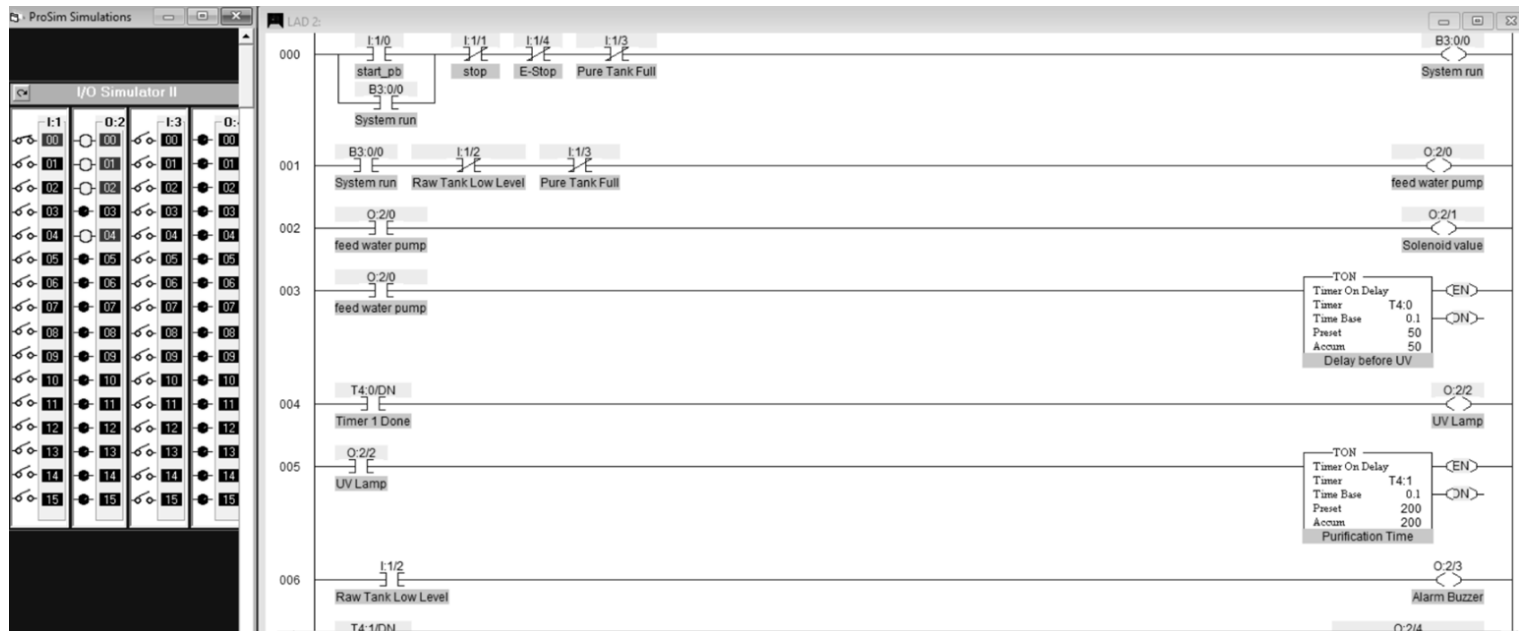
Below is the structured logic equivalent of the automated water purification ladder diagram:

- **Rung 000 (Master Control Latch):** $\text{System_Run (B3:0/0)} = [\text{Start_PB (I:1/0)} \text{ OR } \text{System_Run (B3:0/0)}] \text{ AND NOT Stop (I:1/1) AND NOT E-Stop (I:1/4) AND NOT Pure_Tank_Full (I:1/3)}$
- **Rung 001 (Pump Activation):** $\text{Feed_Water_Pump (O:2/0)} = \text{System_Run (B3:0/0)} \text{ AND NOT Raw_Tank_Low_Level (I:1/2) AND NOT Pure_Tank_Full (I:1/3)}$
- **Rung 002 (Valve Control):** $\text{Solenoid_Valve (O:2/1)} = \text{Feed_Water_Pump (O:2/0)}$
- **Rung 003 (UV Delay Timer):** $\text{Timer_1 (T4:0)} = \text{TON(Enable: Feed_Water_Pump (O:2/0), Time Base: 0.1, Preset: 50)}$
- **Rung 004 (UV Sterilization):** $\text{UV_Lamp (O:2/2)} = \text{Timer_1_Done (T4:0/DN)}$
- **Rung 005 (Purification Cycle Timer):** $\text{Timer_2 (T4:1)} = \text{TON(Enable: UV_Lamp (O:2/2), Time Base: 0.1, Preset: 200)}$
- **Rung 006 (Fault Alarm):** $\text{Alarm_Buzzer (O:2/3)} = \text{Raw_Tank_Low_Level (I:1/2)}$
- **Rung 007 (Cycle Completion Indicator):** $\text{Purification_Complete_Ind (O:2/4)} = \text{Timer_2_Done (T4:1/DN)}$

4.5 OUTPUT ANALYSIS

The output analysis of the automated water purification system was conducted by monitoring the virtual I/O states within the LogixPro and Twindosvite simulation environments. By forcing various input conditions and observing the corresponding output responses, the system's reliability and adherence to the design objectives were validated. The analysis of the system's behavior under different operating conditions is summarized below:

- **Normal Cycle Execution:** Upon activating the Start push-button (I:1/0) under healthy tank conditions, the master latch was successfully established. The simulation confirmed that the Feed Water Pump (O:2/0) and Solenoid Valve (O:2/1) energized simultaneously. The Timer On Delay (T4:0) accurately held the UV Lamp (O:2/2) in an off state for exactly 5 seconds, validating the required flow-stabilization buffer. Following this delay, the UV Lamp activated, and the main purification timer precisely tracked the 20-second (200 preset at 0.1 time base) treatment cycle before triggering the Purification Completion Indicator (O:2/4).
- **Dry-Run Protection (Raw Tank Empty):** To test the hardware protection logic, the Raw Tank Low Level sensor (I:1/2) was manually forced high during an active cycle. The system immediately dropped the Feed Water Pump and Solenoid Valve outputs, successfully preventing a simulated dry-run scenario. Simultaneously, the Alarm Buzzer (O:2/3) output energized, confirming that the operator would receive an immediate auditory warning of the fault.
- **Overflow Prevention (Pure Tank Full):** When the Pure Tank Full sensor (I:1/3) was toggled to simulate a full holding tank, the master control rung instantly broke continuity. All active outputs, including the pump and UV lamp, were safely de-energized. This confirmed that the automated interlocks successfully eliminate the risk of facility flooding and wasted purified water without requiring human intervention.
- **Emergency Stop Override:** Activating the E-Stop (I:1/4) input instantly unlatched the System Run bit (B3:0/0), effectively overriding all ongoing timers and active outputs. The system remained safely halted until the E-Stop was reset and the Start button was manually pressed again.



CHAPTER 5: CONCLUSION

This eight-week summer training provided practical, hands-on exposure to the design and implementation of PLC-based automation systems, using an automated water purification system as a representative industrial case study. Beginning with an overview of raw water intake, tank level monitoring, pumping mechanisms, and UV sterilization stages, the training progressively built up practical skills in ladder logic programming, hardware interlock design, timer sequencing, and system validation using simulation software like LogixPro and Twindosvite.

The mini-project undertaken – a PLC-driven automated water purification system – demonstrated that a logic-controlled, closed-loop approach can maintain precise control over fluid processes, execute a consistent and repeatable sterilization delay sequence, and provide fail-safe protection against dry-running and tank overflows, all of which would be difficult to achieve reliably through manual operation alone. At the same time, the project also highlighted the practical challenges involved in automating a continuous flow process, particularly around designing secure "seal-in" latch circuits, synchronizing timer delays, and integrating multiple float sensor interlocks without conflicting states, as discussed in Section 3.7.

Overall, the training reinforced the value of industrial automation in improving consistency, equipment protection, operational efficiency, and operator safety in process industries such as water treatment, and provided a strong practical foundation for further study of PLC programming, software simulation environments, and modern industrial control concepts.

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